

# Assignment 3: Naive Bayes Classifier for Personal Loan Prediction

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2025-10-12

## Data Loading and Exploration

```
data <- read.csv("UniversalBank(1).csv")
str(data)
```

```
## 'data.frame':    5000 obs. of  14 variables:
## $ ID             : int  1 2 3 4 5 6 7 8 9 10 ...
## $ Age            : int  25 45 39 35 35 37 53 50 35 34 ...
## $ Experience      : int  1 19 15 9 8 13 27 24 10 9 ...
## $ Income          : int  49 34 11 100 45 29 72 22 81 180 ...
## $ ZIP.Code        : int  91107 90089 94720 94112 91330 92121 91711 93943 90089 93023
## ...
## $ Family          : int  4 3 1 1 4 4 2 1 3 1 ...
## $ CCAvg           : num  1.6 1.5 1 2.7 1 0.4 1.5 0.3 0.6 8.9 ...
## $ Education       : int  1 1 1 2 2 2 2 3 2 3 ...
## $ Mortgage        : int  0 0 0 0 0 155 0 0 104 0 ...
## $ Personal.Loan    : int  0 0 0 0 0 0 0 0 0 1 ...
## $ Securities.Account: int  1 1 0 0 0 0 0 0 0 0 ...
## $ CD.Account       : int  0 0 0 0 0 0 0 0 0 0 ...
## $ Online           : int  0 0 0 0 0 1 1 0 1 0 ...
## $ CreditCard       : int  0 0 0 0 1 0 0 1 0 0 ...
```

```
head(data)
```

```
##   ID Age Experience Income ZIP.Code Family CCAvg Education Mortgage
## 1  1  25          1     49   91107      4   1.6          1         0
## 2  2  45          19     34   90089      3   1.5          1         0
## 3  3  39          15     11   94720      1   1.0          1         0
## 4  4  35           9    100   94112      1   2.7          2         0
## 5  5  35           8     45   91330      4   1.0          2         0
## 6  6  37          13     29   92121      4   0.4          2        155
##   Personal.Loan Securities.Account CD.Account Online CreditCard
## 1              0                  1          0      0         0
## 2              0                  1          0      0         0
## 3              0                  0          0      0         0
## 4              0                  0          0      0         0
## 5              0                  0          0      0         1
## 6              0                  0          0      1         0
```

```
summary(data)
```

```
##           ID           Age           Experience           Income           ZIP.Code
## Min.      : 1   Min.    :23.00   Min.     :-3.0   Min.      : 8.00   Min.      : 9307
## 1st Qu.:1251   1st Qu.:35.00   1st Qu.:10.0   1st Qu.: 39.00   1st Qu.:91911
## Median :2500   Median :45.00   Median :20.0   Median : 64.00   Median :93437
## Mean    :2500   Mean    :45.34   Mean    :20.1   Mean    : 73.77   Mean    :93153
## 3rd Qu.:3750   3rd Qu.:55.00   3rd Qu.:30.0   3rd Qu.: 98.00   3rd Qu.:94608
## Max.    :5000   Max.    :67.00   Max.    :43.0   Max.    :224.00   Max.    :96651
##           Family           CCAvg           Education           Mortgage
## Min.      :1.000   Min.     : 0.000   Min.     :1.000   Min.      : 0.0
## 1st Qu.:1.000   1st Qu.: 0.700   1st Qu.:1.000   1st Qu.: 0.0
## Median :2.000   Median : 1.500   Median :2.000   Median : 0.0
## Mean    :2.396   Mean    : 1.938   Mean    :1.881   Mean    : 56.5
## 3rd Qu.:3.000   3rd Qu.: 2.500   3rd Qu.:3.000   3rd Qu.:101.0
## Max.    :4.000   Max.    :10.000   Max.     :3.000   Max.     :635.0
## Personal.Loan   Securities.Account   CD.Account           Online
## Min.      :0.000   Min.     :0.0000   Min.     :0.0000   Min.     :0.0000
## 1st Qu.:0.000   1st Qu.:0.0000   1st Qu.:0.0000   1st Qu.:0.0000
## Median :0.000   Median :0.0000   Median :0.0000   Median :1.0000
## Mean    :0.096   Mean    :0.1044   Mean    :0.0604   Mean    :0.5968
## 3rd Qu.:0.000   3rd Qu.:0.0000   3rd Qu.:0.0000   3rd Qu.:1.0000
## Max.    :1.000   Max.     :1.0000   Max.     :1.0000   Max.     :1.0000
##           CreditCard
## Min.      :0.000
## 1st Qu.:0.000
## Median :0.000
## Mean    :0.294
## 3rd Qu.:1.000
## Max.    :1.000
```

## Data Preprocessing

```
# Remove ID and ZIP Code
data <- data[, -c(1, 5)]

# Convert categorical variables to factors
data$Personal.Loan <- as.factor(data$Personal.Loan)
data$Online <- as.factor(data$Online)
data$CreditCard <- as.factor(data$CreditCard)

# Check for missing values
cat("Missing values:", sum(is.na(data)), "\n")
```

```
## Missing values: 0
```

## Partition Data into Training (60%) and Validation

# (40%)

```
set.seed(123)
trainIndex <- sample(1:nrow(data), 0.6 * nrow(data))
train <- data[trainIndex, ]
validation <- data[-trainIndex, ]

cat("Training set size:", nrow(train), "\n")
```

```
## Training set size: 3000
```

```
cat("Validation set size:", nrow(validation), "\n")
```

```
## Validation set size: 2000
```

## A. Create Pivot Table with Online, CC, and Loan

```
# Create pivot table: Online (columns), CC (rows), Loan (secondary rows), Count (values)
pivot_table <- table(train$CreditCard, train$Online, train$Personal.Loan)
names(dimnames(pivot_table)) <- c("CreditCard", "Online", "Personal.Loan")

cat("Pivot Table - Count of customers:\n")
```

```
## Pivot Table - Count of customers:
```

```
cat("Personal.Loan = 0 (Did not accept loan):\n")
```

```
## Personal.Loan = 0 (Did not accept loan):
```

```
print(pivot_table[,,"0"])
```

```
##           Online
## CreditCard  0    1
##           0  785 1145
##           1  317  475
```

```
cat("\nPersonal.Loan = 1 (Accepted loan):\n")
```

```
##
## Personal.Loan = 1 (Accepted loan):
```

```
print(pivot_table[,,"1"])
```

```
##           Online
## CreditCard  0    1
##           0  65 122
##           1  34  57
```

```
# Alternative flat table format
pivot_flat <- ftable(train$Personal.Loan, train$CreditCard, train$Online)
cat("\nFlat Pivot Table:\n")
```

```
##
## Flat Pivot Table:
```

```
print(pivot_flat)
```

```
##           0    1
##
## 0 0    785 1145
##  1    317  475
## 1 0     65  122
##  1     34   57
```

## B. Probability $P(\text{Loan} = 1 \mid \text{CC} = 1, \text{Online} = 1)$

```
# From pivot table: customers with CC=1 and Online=1
cc1_online1_loan1 <- pivot_table["1", "1", "1"]
cc1_online1_loan0 <- pivot_table["1", "1", "0"]
cc1_online1_total <- cc1_online1_loan1 + cc1_online1_loan0

prob_loan_given_cc_online <- cc1_online1_loan1 / cc1_online1_total

cat("Customers with CC=1 and Online=1 who accepted loan:", cc1_online1_loan1, "\n")
```

```
## Customers with CC=1 and Online=1 who accepted loan: 57
```

```
cat("Total customers with CC=1 and Online=1:", cc1_online1_total, "\n")
```

```
## Total customers with CC=1 and Online=1: 532
```

```
cat("P(Loan = 1 | CC = 1, Online = 1) =", round(prob_loan_given_cc_online, 4), "\n")
```

```
## P(Loan = 1 | CC = 1, Online = 1) = 0.1071
```

## C. Create Two Separate Pivot Tables

```
# Pivot table: Loan (rows) by Online (columns)
pivot_loan_online <- table(train$Personal.Loan, train$Online)
rownames(pivot_loan_online) <- c("Loan=0", "Loan=1")
colnames(pivot_loan_online) <- c("Online=0", "Online=1")

cat("Pivot Table - Loan by Online:\n")
```

```
## Pivot Table - Loan by Online:
```

```
print(pivot_loan_online)
```

```
##
##           Online=0 Online=1
## Loan=0         1102      1620
## Loan=1           99       179
```

```
# Pivot table: Loan (rows) by CC (columns)
pivot_loan_cc <- table(train$Personal.Loan, train$CreditCard)
rownames(pivot_loan_cc) <- c("Loan=0", "Loan=1")
colnames(pivot_loan_cc) <- c("CC=0", "CC=1")

cat("\nPivot Table - Loan by CreditCard:\n")
```

```
##
## Pivot Table - Loan by CreditCard:
```

```
print(pivot_loan_cc)
```

```
##
##           CC=0 CC=1
## Loan=0  1930  792
## Loan=1   187   91
```

## D. Compute Required Probabilities

```
# Extract counts from pivot tables
loan1_total <- sum(train$Personal.Loan == "1")
loan0_total <- sum(train$Personal.Loan == "0")
total_customers <- nrow(train)

# i.  $P(CC = 1 \mid Loan = 1)$ 
cc1_loan1 <- pivot_loan_cc["Loan=1", "CC=1"]
p_cc1_given_loan1 <- cc1_loan1 / loan1_total

# ii.  $P(Online = 1 \mid Loan = 1)$ 
online1_loan1 <- pivot_loan_online["Loan=1", "Online=1"]
p_online1_given_loan1 <- online1_loan1 / loan1_total

# iii.  $P(Loan = 1)$ 
p_loan1 <- loan1_total / total_customers

# iv.  $P(CC = 1 \mid Loan = 0)$ 
cc1_loan0 <- pivot_loan_cc["Loan=0", "CC=1"]
p_cc1_given_loan0 <- cc1_loan0 / loan0_total

# v.  $P(Online = 1 \mid Loan = 0)$ 
online1_loan0 <- pivot_loan_online["Loan=0", "Online=1"]
p_online1_given_loan0 <- online1_loan0 / loan0_total

# vi.  $P(Loan = 0)$ 
p_loan0 <- loan0_total / total_customers

cat("i.    $P(CC = 1 \mid Loan = 1) =$ ", round(p_cc1_given_loan1, 4), "\n")
```

```
## i.    $P(CC = 1 \mid Loan = 1) = 0.3273$ 
```

```
cat("ii.   $P(Online = 1 \mid Loan = 1) =$ ", round(p_online1_given_loan1, 4), "\n")
```

```
## ii.   $P(Online = 1 \mid Loan = 1) = 0.6439$ 
```

```
cat("iii.  $P(Loan = 1) =$ ", round(p_loan1, 4), "\n")
```

```
## iii.  $P(Loan = 1) = 0.0927$ 
```

```
cat("iv.   $P(CC = 1 \mid Loan = 0) =$ ", round(p_cc1_given_loan0, 4), "\n")
```

```
## iv.   $P(CC = 1 \mid Loan = 0) = 0.291$ 
```

```
cat("v.    $P(Online = 1 \mid Loan = 0) =$ ", round(p_online1_given_loan0, 4), "\n")
```

```
## v.    $P(Online = 1 \mid Loan = 0) = 0.5952$ 
```

```
cat("vi. P(Loan = 0) =", round(p_loan0, 4), "\n")
```

```
## vi. P(Loan = 0) = 0.9073
```

## E. Compute Naive Bayes Probability $P(\text{Loan} = 1 \mid \text{CC} = 1, \text{Online} = 1)$

```
# Using Bayes theorem:
#  $P(\text{Loan}=1 \mid \text{CC}=1, \text{Online}=1) = P(\text{CC}=1 \mid \text{Loan}=1) * P(\text{Online}=1 \mid \text{Loan}=1) * P(\text{Loan}=1) / P(\text{CC}=1, \text{Online}=1)$ 
#
# For denominator:  $P(\text{CC}=1, \text{Online}=1) = P(\text{CC}=1, \text{Online}=1 \mid \text{Loan}=1) * P(\text{Loan}=1) + P(\text{CC}=1, \text{Online}=1 \mid \text{Loan}=0) * P(\text{Loan}=0)$ 
# Assuming independence:  $P(\text{CC}=1, \text{Online}=1 \mid \text{Loan}=1) = P(\text{CC}=1 \mid \text{Loan}=1) * P(\text{Online}=1 \mid \text{Loan}=1)$ 

# Numerator
numerator <- p_cc1_given_loan1 * p_online1_given_loan1 * p_loan1

# Denominator
prob_cc1_online1_given_loan1 <- p_cc1_given_loan1 * p_online1_given_loan1
prob_cc1_online1_given_loan0 <- p_cc1_given_loan0 * p_online1_given_loan0
denominator <- (prob_cc1_online1_given_loan1 * p_loan1) + (prob_cc1_online1_given_loan0 * p_loan0)

# Naive Bayes probability
nb_probability <- numerator / denominator

cat("Naive Bayes Calculation:\n")
```

```
## Naive Bayes Calculation:
```

```
cat("Numerator:  $P(\text{CC}=1 \mid \text{Loan}=1) * P(\text{Online}=1 \mid \text{Loan}=1) * P(\text{Loan}=1)$  =", round(numerator, 6), "\n")
```

```
## Numerator:  $P(\text{CC}=1 \mid \text{Loan}=1) * P(\text{Online}=1 \mid \text{Loan}=1) * P(\text{Loan}=1) = 0.019531$ 
```

```
cat("Denominator =", round(denominator, 6), "\n")
```

```
## Denominator = 0.176651
```

```
cat("\nP(Loan = 1 | CC = 1, Online = 1) =", round(nb_probability, 4), "\n")
```

```
##
##  $P(\text{Loan} = 1 \mid \text{CC} = 1, \text{Online} = 1) = 0.1106$ 
```

## F. Compare with Pivot Table Result from (B)

```
# Comparison
comparison_df <- data.frame(
  Method = c("Pivot Table (B)", "Naive Bayes (E)"),
  Probability = c(prob_loan_given_cc_online, nb_probability)
)
print(comparison_df)
```

```
##           Method Probability
## 1 Pivot Table (B)    0.1071429
## 2 Naive Bayes (E)    0.1105637
```

```
cat("\nDifference:", round(abs(prob_loan_given_cc_online - nb_probability), 4), "\n")
```

```
##
## Difference: 0.0034
```

```
cat("\nThe pivot table gives the exact probability from observed data, while Naive Bayes")
```

```
##
## The pivot table gives the exact probability from observed data, while Naive Bayes
```

```
cat("\nassumes conditional independence. The pivot table is more accurate for this specific")
```

```
##
## assumes conditional independence. The pivot table is more accurate for this specific
```

```
cat("\ncombination, but Naive Bayes generalizes better to new data.\n")
```

```
##
## combination, but Naive Bayes generalizes better to new data.
```

## G. Run Naive Bayes Model and Compare

```
# Build Naive Bayes model using only Online and CreditCard predictors
nb_model <- naiveBayes(Personal.Loan ~ Online + CreditCard, data = train)
print(nb_model)
```



```
##
## Naive Bayes Classifier for Discrete Predictors
##
## Call:
## naiveBayes.default(x = X, y = Y, laplace = laplace)
##
## A-priori probabilities:
## Y
##           0           1
## 0.90733333 0.09266667
##
## Conditional probabilities:
##   Online
## Y           0           1
## 0 0.4048494 0.5951506
## 1 0.3561151 0.6438849
##
##   CreditCard
## Y           0           1
## 0 0.7090375 0.2909625
## 1 0.6726619 0.3273381
```

```
# Predict for customer with CC=1 and Online=1
test_customer <- data.frame(Online = factor("1", levels = c("0", "1")),
                           CreditCard = factor("1", levels = c("0", "1")))

prediction <- predict(nb_model, test_customer, type = "raw")
cat("\n\nModel prediction for customer with CC=1 and Online=1:\n")
```

```
##
##
## Model prediction for customer with CC=1 and Online=1:
```

```
cat("P(Loan = 0 | CC = 1, Online = 1) =", round(prediction[1], 4), "\n")
```

```
## P(Loan = 0 | CC = 1, Online = 1) = 0.8894
```

```
cat("P(Loan = 1 | CC = 1, Online = 1) =", round(prediction[2], 4), "\n")
```

```
## P(Loan = 1 | CC = 1, Online = 1) = 0.1106
```

```
cat("\nComparison:\n")
```

```
##
## Comparison:
```

```
cat("Manually calculated (E):", round(nb_probability, 4), "\n")
```

```
## Manually calculated (E): 0.1106
```

```
cat("Model output:", round(prediction[2], 4), "\n")
```

```
## Model output: 0.1106
```

```
cat("Difference:", round(abs(nb_probability - prediction[2]), 6), "\n\n")
```

```
## Difference: 0
```

```
cat("The values match closely, confirming our manual calculation.\n")
```

```
## The values match closely, confirming our manual calculation.
```

## Additional: Full Model with All Predictors

```
# Build full Naive Bayes model with all predictors
nb_full <- naiveBayes(Personal.Loan ~ ., data = train)

# Predictions on validation set
pred_val <- predict(nb_full, validation)

# Confusion matrix
conf_matrix <- confusionMatrix(pred_val, validation$Personal.Loan, positive = "1")
print(conf_matrix)
```

```
## Confusion Matrix and Statistics
##
##           Reference
## Prediction    0    1
##           0 1629   86
##           1  169  116
##
##           Accuracy : 0.8725
##           95% CI : (0.8571, 0.8868)
##    No Information Rate : 0.899
##    P-Value [Acc > NIR] : 0.9999
##
##           Kappa : 0.4062
##
##    McNemar's Test P-Value : 2.821e-07
##
##           Sensitivity : 0.5743
##           Specificity : 0.9060
##    Pos Pred Value : 0.4070
##    Neg Pred Value : 0.9499
##           Prevalence : 0.1010
##    Detection Rate : 0.0580
##    Detection Prevalence : 0.1425
##    Balanced Accuracy : 0.7401
##
##           'Positive' Class : 1
##
```

```
cat("\nModel Performance:\n")
```

```
##
## Model Performance:
```

```
cat("Accuracy:", round(conf_matrix$overall['Accuracy'], 4), "\n")
```

```
## Accuracy: 0.8725
```

```
cat("Sensitivity (Recall):", round(conf_matrix$byClass['Sensitivity'], 4), "\n")
```

```
## Sensitivity (Recall): 0.5743
```

```
cat("Specificity:", round(conf_matrix$byClass['Specificity'], 4), "\n")
```

```
## Specificity: 0.906
```

```
cat("Precision:", round(conf_matrix$byClass['Precision'], 4), "\n")
```

```
## Precision: 0.407
```

The Naive Bayes classifier successfully predicts personal loan acceptance with good accuracy. The manual calculations match the model output, confirming the correctness of the Naive Bayes approach. The key findings are:

1. Pivot Table Probability (B): More accurate for exact combinations seen in training data
2. Naive Bayes Probability (E): Uses independence assumption, generalizes better to unseen data
3. Model Performance: The full model achieves high accuracy on validation data
4. The conditional independence assumption works reasonably well for this dataset, though there is a small difference between exact probability and Naive Bayes estimate.